Lecture 10 - Product/Quotient Rule, Trig Derivatives

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Exam 1 Recap

1. All grades for Exam 1 released, regrade requests open until next Monday.
2. After Monday, curve will be determined
3. Recap: Essential Discontinuity, Vertical Tangent Singularity, Vertical Asymptotes

Also, Anonymous Feedback Form.

Quiz Time!

Let’s get into neat rows, spread out, and slay this quiz :)

Product and Quotient Rule

Long Term Goals

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- Trigonometric Rules: $\frac{d}{dx}\sin(x)$ etc.
- Exponential/Logarithm Rules: $\frac{d}{dx}e^x$, $\frac{d}{dx}\ln(x)$

Combination Rules:

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Combination Rules:

- Sum/Difference Rule: $\frac{d}{dx}(f(x) \pm g(x))$
- Product Rule: $\frac{d}{dx}f(x)g(x)$
- Chain Rule: $\frac{d}{dx}g(f(x))$
- Constant Multiple Rule: $\frac{d}{dx}c \cdot f(x)$
- Quotient Rule: $\frac{d}{dx}\frac{f(x)}{g(x)}$
- Inverse Function Rule: $\frac{d}{dx}f^{-1}(x)$

Product Rule I

We want to determine what $\frac{d}{dx}(f(x) \cdot g(x))$ is, hopefully in terms of simpler derivatives involving $f(x), g(x)$.

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Is it just $f'(x)g'(x)$? NO NO NO NO NO!



Figure: My one wish willow is for you to not compute derivates of products like this!

Product Rule II

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Linearization of $f(x)$ at $x = x_0$: $f(x_0) + f'(x_0)(x - x_0)$,
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Linearization of $f(x)g(x)$ at $x = x_0$:

$$f(x)g(x) \approx (f(x_0) + f'(x_0)(x - x_0)) \cdot (g(x_0) + g'(x_0)(x - x_0)) \tag{1}$$

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$$f(x)g(x) \approx (f(x_0) + f'(x_0)(x - x_0)) \cdot (g(x_0) + g'(x_0)(x - x_0)) \quad (1)$$

$$= f(x_0)g(x_0) + f'(x_0)g(x_0)(x - x_0) + f(x_0)g'(x_0)(x - x_0) + f'(x_0)g'(x_0)(x - x_0)^2 \quad (2)$$

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So why is it NOT $f'(x)g'(x)$?

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So why is it NOT $f'(x)g'(x)$? Because that's the coefficient of something involving a *quadratic* term, not linear!

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So why is it NOT $f'(x)g'(x)$? Because that's the coefficient of something involving a *quadratic* term, not linear!
The derivative would be the slope on the *linear* term.

Product Rule III

The Product Rule

$$\frac{d}{dx}(f(x) \cdot g(x)) = f'(x)g(x) + f(x)g'(x) \tag{4}$$

Product Rule Examples I

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$$x^2 e^x + \frac{2^x}{x}? \tag{5}$$

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$$= (\frac{d}{dx}x^2) \cdot (e^x) + (x^2) \cdot (\frac{d}{dx}e^x) + \frac{d}{dx}(2^x \cdot \frac{1}{x}) \tag{7}$$

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$$= (2x) \cdot e^x + x^2 \cdot e^x + (\frac{d}{dx}2^x) \cdot (\frac{1}{x}) + (2^x) \cdot (\frac{d}{dx}\frac{1}{x}) \tag{8}$$

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$$= 2xe^x + x^2 e^x + \ln(2)2^x \cdot \frac{1}{x} + 2^x \cdot \frac{-1}{x^2} \tag{9}$$

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Suppose we've measured the relevant quantities at:

$$m(1) = 50mi, \quad e(1) = \frac{1}{25}\frac{gal}{mi}, \quad p(1) = 4.15\frac{\$}{gal} \tag{10}$$

$$m'(1) = 0\frac{mi}{day}, \quad e'(1) = \frac{1}{1000}\frac{\frac{gal}{mi}}{day}, \quad p'(1) = 0.01\frac{\frac{\$}{gal}}{day} \tag{11}$$

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Product Rule Examples III

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$$= 0 \cdot \frac{1}{25}\frac{gal}{mi} \cdot 4.15\frac{\$}{gal} \tag{16}$$

$$+ 50mi \cdot \frac{1}{1000}\frac{\frac{gal}{mi}}{day} \cdot 4.15\frac{\$}{gal} \tag{17}$$

$$+ 50mi \cdot \frac{1}{25}\frac{gal}{mi} \cdot 0.01\frac{\frac{\$}{gal}}{day} \tag{18}$$

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Similarly, we'd want to determine $\frac{d}{dx}\frac{f(x)}{g(x)}$, again in terms of simpler derivatives involving $f(x)$, $g(x)$.

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Is it just $\frac{f'(x)}{g'(x)}$? SUPER NO! NO! NO!



Figure: I'm amending my one wish willow to also include you not computing derivatives of quotients like this.

Quotient Rule II

We can determine what it actually is by realizing that:

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First, we note that for the derivative of $\frac{f(x)}{g(x)}$ at $x = x_0$ to exist, the function $\frac{f(x)}{g(x)}$ must be continuous at $x = x_0$, so $g(x_0) \neq 0$.

This allows us to approximate $\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)}$.

Quotient Rule III

Since

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$$\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)} = \frac{1}{(g(x_0) - g'(x_0)x_0) + g'(x_0)x} \tag{22}$$

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we can focus on understanding how to linearize functions like

$$\frac{1}{a + bx}. \tag{23}$$

We compute:

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We compute:

$$\begin{aligned} \frac{d}{dx}\frac{1}{a+bx} &= \frac{d}{dx}\frac{1}{b}\frac{1}{x + \frac{a}{b}} = \frac{1}{b}\frac{d}{dx}\frac{1}{x + \frac{a}{b}} \\ &= \frac{1}{b}\frac{-1}{(x + \frac{a}{b})^2} \end{aligned} \tag{24}$$

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Since

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$$\begin{aligned} \frac{d}{dx}\frac{1}{a+bx} &= \frac{d}{dx}\frac{1}{b}\frac{1}{x+\frac{a}{b}} = \frac{1}{b}\frac{d}{dx}\frac{1}{x+\frac{a}{b}} \\ &= \frac{1}{b}\frac{-1}{(x+\frac{a}{b})^2} = \frac{1}{b}\frac{-1}{x^2+\frac{2a}{b}x+\frac{a^2}{b^2}} \end{aligned} \tag{24}$$

Quotient Rule III

Since

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0) + g'(x_0)(x - x_0)} = \frac{1}{(g(x_0) - g'(x_0)x_0) + g'(x_0)x} \tag{22}$$

we can focus on understanding how to linearize functions like

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And so,

$$\frac{1}{g(x)} \approx \frac{1}{g(x_0)} + \frac{-g'(x_0)}{g(x_0)^2}(x - x_0) \tag{27}$$

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Quotient Rule V

Moral of the story: it's HARD to actually get the derivative of a quotient

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E.g. computing:

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$$= \frac{(2e^x + 2xe^x + 6x) \cdot (x^3 + 2x + 1) - (2xe^x + 3x^2) \cdot (3x^2 + 2)}{(x^3 + 2x + 1)^2} \tag{36}$$

Trigonometric Derivatives

Derivative of $\sin(x)$ I

Recall from our discussion on the Squeeze Theorem:

$$\lim_{x \to 0} \frac{\sin(x)}{x} = 1 \tag{37}$$

$$\lim_{x \to 0} \frac{\cos(x) - 1}{x} = 0 \tag{38}$$

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What is this limit in terms of derivatives?

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What is this limit in terms of derivatives?

$$\frac{d}{dx} |_{x=0} \sin(x) = 1, \frac{d}{dx} |_{x=0} \cos(x) = 0 \tag{39}$$

Derivative of $\sin(x)$ II

What about the derivative function?

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Derivative of $\sin(2x)$

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$$\frac{d}{dx}\sin(2x) = \frac{d}{dx}(2\sin(x)\cos(x)) \tag{56}$$

$$= 2\frac{d}{dx}(\sin(x)\cos(x)) \tag{57}$$

$$= 2 \cdot \left(\left(\frac{d}{dx}\sin(x)\right) \cdot \cos(x) + \sin(x) \cdot \left(\frac{d}{dx}\cos(x)\right)\right) \tag{58}$$

$$= 2 \cdot (\cos(x) \cdot \cos(x) + \sin(x) \cdot (-\sin(x))) \tag{59}$$

$$= 2 \cdot (\cos(x)^2 - \sin(x)^2) \tag{60}$$

$$= 2 \cdot \cos(2x) \tag{61}$$

Worksheet Time!

Let's get into groups (or if you really like working alone, please stay so I can be here to help :)) and work on the worksheet!

End of Class!

That’s all folks!